

A Simple Metropolis–Hastings Example for a Bimodal Target Distribution

Target distribution

Consider the target distribution

$$f(x) = \frac{1}{2}\mathcal{N}(-2, 1) + \frac{1}{2}\mathcal{N}(2, 1), \quad (1)$$

where each Gaussian has variance equal to one. Explicitly,

$$f(x) = \frac{1}{2\sqrt{2\pi}} \left[\exp\left(-\frac{(x+2)^2}{2}\right) + \exp\left(-\frac{(x-2)^2}{2}\right) \right]. \quad (2)$$

This is a bimodal distribution with two peaks, one centered at $x = -2$ and the other at $x = 2$.

Proposal distribution

Assume that the current point of the Markov chain is

$$x_t = 0.6. \quad (3)$$

We use a Gaussian proposal distribution centered at the current point,

$$q(x'|x_t) = \mathcal{N}(x_t, \sigma^2), \quad (4)$$

with

$$\sigma^2 = 0.4, \quad \sigma = \sqrt{0.4} \simeq 0.632. \quad (5)$$

Therefore, the proposed point is generated as

$$x' = x_t + \eta, \quad \eta \sim \mathcal{N}(0, 0.4), \quad (6)$$

or equivalently

$$x' \sim \mathcal{N}(x_t, 0.4). \quad (7)$$

The proposal distribution only suggests a candidate point. The candidate is then accepted or rejected according to the Metropolis–Hastings criterion.

Acceptance probability and uniform random number

The general Metropolis–Hastings acceptance probability is

$$A(x_t \rightarrow x') = \min \left[1, \frac{f(x')q(x_t|x')}{f(x_t)q(x'|x_t)} \right]. \quad (8)$$

In this example the Gaussian proposal is symmetric, because

$$q(x'|x_t) = q(x_t|x'). \quad (9)$$

Therefore, the proposal terms cancel and the acceptance probability becomes

$$A(x_t \rightarrow x') = \min \left[1, \frac{f(x')}{f(x_t)} \right]. \quad (10)$$

To make the actual accept/reject decision, one draws a random number

$$u \sim U(0, 1), \quad (11)$$

and accepts the proposed move if

$$u < A(x_t \rightarrow x'). \quad (12)$$

Thus, the number 1 in the expression $\min(1, \dots)$ only ensures that the acceptance probability cannot exceed one. The random number u is what decides whether a move with $A < 1$ is accepted or rejected.

Move 1: an accepted move

Suppose that, starting from

$$x_t = 0.6, \quad (13)$$

the proposal distribution generates

$$x' = 1.2. \quad (14)$$

The acceptance probability is

$$A(0.6 \rightarrow 1.2) = \min \left[1, \frac{f(1.2)}{f(0.6)} \right]. \quad (15)$$

First, evaluate the target distribution at the current point:

$$f(0.6) = \frac{1}{2\sqrt{2\pi}} \left[\exp \left(-\frac{(0.6+2)^2}{2} \right) + \exp \left(-\frac{(0.6-2)^2}{2} \right) \right] \quad (16)$$

$$= \frac{1}{2\sqrt{2\pi}} \left[\exp \left(-\frac{2.6^2}{2} \right) + \exp \left(-\frac{(-1.4)^2}{2} \right) \right] \quad (17)$$

$$\simeq 0.08166. \quad (18)$$

Next, evaluate the target distribution at the proposed point:

$$f(1.2) = \frac{1}{2\sqrt{2\pi}} \left[\exp \left(-\frac{(1.2+2)^2}{2} \right) + \exp \left(-\frac{(1.2-2)^2}{2} \right) \right] \quad (19)$$

$$= \frac{1}{2\sqrt{2\pi}} \left[\exp \left(-\frac{3.2^2}{2} \right) + \exp \left(-\frac{(-0.8)^2}{2} \right) \right] \quad (20)$$

$$\simeq 0.14604. \quad (21)$$

Therefore,

$$\frac{f(1.2)}{f(0.6)} \simeq \frac{0.14604}{0.08166} \simeq 1.79. \quad (22)$$

Thus,

$$A(0.6 \rightarrow 1.2) = \min(1, 1.79) = 1. \quad (23)$$

Since $A = 1$, any random number $u \in [0, 1]$ satisfies $u < A$. For example, if

$$u = 0.42, \quad (24)$$

then

$$u = 0.42 < A = 1, \quad (25)$$

and the move is accepted. Therefore,

$$x_{t+1} = 1.2. \quad (26)$$

Move 2: a rejected move

Now assume that the chain is at

$$x_t = 1.2, \quad (27)$$

and the proposal distribution generates a new candidate point

$$x' = 0. \quad (28)$$

This proposal is generated from

$$x' \sim \mathcal{N}(1.2, 0.4). \quad (29)$$

The acceptance probability is

$$A(1.2 \rightarrow 0) = \min \left[1, \frac{f(0)}{f(1.2)} \right]. \quad (30)$$

The value of the target distribution at $x = 0$ is

$$f(0) = \frac{1}{2\sqrt{2\pi}} \left[\exp \left(-\frac{(0+2)^2}{2} \right) + \exp \left(-\frac{(0-2)^2}{2} \right) \right] \quad (31)$$

$$= \frac{1}{2\sqrt{2\pi}} \left[\exp \left(-\frac{2^2}{2} \right) + \exp \left(-\frac{(-2)^2}{2} \right) \right] \quad (32)$$

$$\simeq 0.05399. \quad (33)$$

Using the previous result $f(1.2) \simeq 0.14604$, we obtain

$$\frac{f(0)}{f(1.2)} \simeq \frac{0.05399}{0.14604} \simeq 0.37. \quad (34)$$

Therefore,

$$A(1.2 \rightarrow 0) = \min(1, 0.37) = 0.37. \quad (35)$$

Now draw a random number from the uniform distribution. Suppose, for example, that

$$u = 0.70. \quad (36)$$

Since

$$u = 0.70 > A = 0.37, \quad (37)$$

the move is rejected. Therefore, the Markov chain remains at the previous point:

$$x_{t+1} = x_t = 1.2. \quad (38)$$

Summary

The proposal distribution $q(x'|x_t)$ determines which candidate point is suggested. In this example it is a Gaussian distribution centered at the current point, with variance $\sigma^2 = 0.4$.

The acceptance probability determines whether the proposed point is accepted. For a symmetric proposal, it is given by

$$A(x_t \rightarrow x') = \min \left[1, \frac{f(x')}{f(x_t)} \right]. \quad (39)$$

Finally, a random number $u \sim U(0, 1)$ is drawn. If $u < A$, the proposed move is accepted; otherwise, it is rejected.

In the first move, the proposal $0.6 \rightarrow 1.2$ moves toward a higher-probability region of the target distribution, so the acceptance probability is $A = 1$ and the move is always accepted. In the second move, the proposal $1.2 \rightarrow 0$ moves toward the low-probability valley between the two peaks, so the acceptance probability is only $A \simeq 0.37$. With $u = 0.70$, this second move is rejected.